TABLE S12a

GPT-6 ASTRA CROSS-MODEL AGREEMENT CALIBRATION

Clustering	Score	Brier	Log loss	ECE
Leiden	Raw agreement	0.2928	4.3298	0.2111
Leiden	Logistic recalibration	0.2577	0.7166	0.1295
Leiden	Development constant	0.2545	0.7022	0.1023
FlowSOM	Raw agreement	0.3348	4.8255	0.2953
FlowSOM	Logistic recalibration	0.2670	0.7331	0.1717
FlowSOM	Development constant	0.2634	0.7228	0.1325
SpatialSort	Raw agreement	0.3559	4.7292	0.3305
SpatialSort	Logistic recalibration	0.2191	0.6304	0.0803
SpatialSort	Development constant	0.2253	0.6440	0.0585
PIXIE	Raw agreement	0.3116	4.1445	0.3292
PIXIE	Logistic recalibration	0.1987	0.5694	0.1020
PIXIE	Development constant	0.2264	0.6453	0.1408

Lower scores are better. These are equal-test-region metrics on hash-held-out evaluation clusters, not the full benchmark population. Development uses disjoint clusters and regions. Agreement support has denominator four. This is retrospective within-donor reference-agreement calibration, not repeatability, native model confidence or biological certainty.